

REMOTE SENSING ANALYSIS OF LAND RESOURCES FOR PLANNING AND DEVELOPMENT IN AND AROUND PULIVENDULA – SANIVARIPALLI AREA, ANDHRA PRADESH

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INTRODUCTION

The use of remote sensing techniques has become increasingly important in describing a variety of satellite-derived data sets and their application to understand changes in the landscape (Kouchoukos *et al.*, 1997). Remote sensing has long been recommended for its potential to detect, map and monitor degradation problems (Hellden and Stern, 1980; Sabins, 1987; Frederiksen, 1993; Mohammed, 1993; Raina *et al.*, 1993; Tripathy *et al.*, 1996; Sujatha *et al.*, 2000) including their spread and effects with time (Sommer *et al.*, 1998). Use of remotely sensed imagery evolved on the basis that traditional survey become expensive and time-consuming. It is especially useful in areas that are not accessible.

Land use refers to mans activities which are directly related to land where as land cover refers to biophysical materials found on the land use surface (Jensen, 2007). The terms land use and land cover is strongly correlated and in many cases has been used interchangeably. Land use practices of a region are influenced by a number of parameters namely physical and chemical environments, socio-economic factors and needs for the masses. Ever increasing demand due to rapid growth of population has put heavy pressure on natural resources of the country in general and the study is in particular.

The knowledge of land use and land cover is very important for land planning and land management activities (Reddy, 2002). Land is the most important natural resource on which most developmental activities are based. The application of remotely sensed data for land use mapping is one of the most acceptable techniques since 1940. Over the past few decades there has been raising concerns from the researchers and planners about the increasing anthropogenic influence on land resources. More recently high resolution satellite data and aerial photography have enhanced and opened new fields for detailed planning of land use and land cover mapping for various developmental and scientific schemes (Burrough, 1986; Sabins, 1987; Chauhan, 2003).

Multidisciplinary scientific integrated surveys were carried out to quantify the resource potential of the area, to know the status of exploitation of resources and to identify any degradation due to unscientific management. The investigation agents broadly outline the development options based on available resources. The present study aims to study the present status of land resources, water resources, soil productivity, cropping patterns, forest cover etc., using satellites data, collateral data and field data and identifying the problems of environmental resources depletion and causes of depletion and to develop spatial database and non-spatial (attribute) database for various levels of planning and decision support to meet requirements of urban planning and management and governance. The thematic maps produced on resources will enable planners to formulate programs to optimize productivity from existing resources and to initiate measures to correct

ABSTRACT

Anthropogenic changes in land use mostly deforestation and conversion of pastures into agricultural lands are being increasingly recognized as critical factors influencing global change. Land information is essential for making decisions related to land development and is important for the betterment of both rural and urban regions. Conventional ground methods of land use mapping are labour extensive, time consuming and are relatively done infrequently. Remote Sensing (RS), integrated with Geographic information system (GIS), provides an effective tool for analysis of land use and land cover changes at a regional level. The geospatial technology of RS and GIS holds the potential for timely and cost-effective assessment of natural resources. A total of three thematic maps such as land use and land cover, drainage and slope maps were prepared based on image interpretation studies with limited field checks. The land use-land cover pattern fall under the broad categories of built-up land, agricultural land, forest land and uncultivable land. The most of the area have uncultivable barren and waste land which is unused for neither cultivation nor settlements and is suggested for industrialization and urban planning. The forest land is much less than the required percentage according to the forest department to maintain the eco-balance in the region.

KEY WORDS

Anthropogenic changes
Land use management
Remote Sensing, Geographic information system, Pulivendula, Sanivaripalli

Received : 19.02.2011

Revised : 24.04.2011

Accepted : 27.06.2011

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imbalances due to unscientific management and inherent deficiency.

Location accessibility

The area lies between Pulivendula and Sanivaripalli situated between parallels of 78°00'0" to 78°15'0" E L and 14°15' 0" to 14°30'0"N L with intended boundary falling in Survey of India (SOI) topographic sheet # 57J 03 on 1:50,000 scale covering an of over 720 km². The study area includes five revenue mandals namely, Pulivendula, Lingala, Udumakurti, Krishnamgaripalli, and Sanivaripalli and is shown in Fig. 1.



Figure 1: Map showing location of the study area

Physiography and climate

The central and southern part of study area is occupied by high hills, ridges and valleys. The value of contours in the study area ranges from 200-600 m. The slope category is within 1-15 degrees. The Maddaleru is the only river in the study area which is a seasonal river fed by monsoon (June – October), which is flowing in the NW direction in the southern part of the study area. Some major tanks also cover in the same area. The climate is dry with mean annual rainfall of 100-150 cm and mean annual temperature of 32°C. The month of May is considered to be the hottest (45°C) while December is considered to be the coldest (25°C). The length of any crop is over 120 days.

Geology

The study area exposes the rocks belonging to Peninsular Gneissic Complex of Archean and Papaghni (Gulcheru and Vempalli formations), Chitravathi group of Cuddapah Super group. The Gulcheru formation is represented by basal polymictic conglomerate/grit, arkose, quartzite with intercalations of shale and overlies the Archean to Paleoproterozoic basement consisting of granite/ gneiss and narrow linear schist belts with a profound Eparchean unconformity (Rao *et al.*, 1987). The arenaceous sequence of Gulcheru quartzite is successively overlain by dominantly calcareous rocks of Vempalli formation having lithounits of stromatolitic dolomites at places, shale, mudstone and chert. Basaltic lava flows and dolerite sills/dykes are present in this formation. A number of transverse faults and fractures affecting both the basement and the overlying sediments are basement rooted and subsequently reactivated.

MATERIALS AND METHODS

Data used and methodology

The study used a satellite geocoded imagery (IRS P6 LISS-III), acquired June 2001 with spatial resolution of 30 meters, Survey of India toposheet 57 J 03 on 1:50,000 scale and tracing sheets. The digital data were processed for geometric and radiometric corrections using ERDAS image processing software. A number of softwares such as Arc GIS version 9.0, ERDAS Imagine version 9.1 and Microsoft office processing tools (Word, Excel) are used in the study. For processing ERDAS was considered useful. The study was applied both visual and digital Landsat image interpretation in order to generate various thematic maps respectively. The field equipments and tools used for fieldwork include Brunton compass, Garmin Trex H (GPS) receiver, slope meter, digital camera, soil knife, hammer, and measuring tape. The main methodology of the research is to prepare thematic maps with the application of remote sensing and GIS software. The actual work involved a comprehensive gathering of information and linked to the related activities. The development and planning of such a vast area for a proposed project requires various data information from the field to identify the terrain characteristics, etc. This is composed of getting use of space images and GIS, complemented with truthing fieldwork as well as making use of the available information from several concerned agencies and sectors in the study area. The flow chart showing the methodology of land resources analysis is given in Fig. 2.

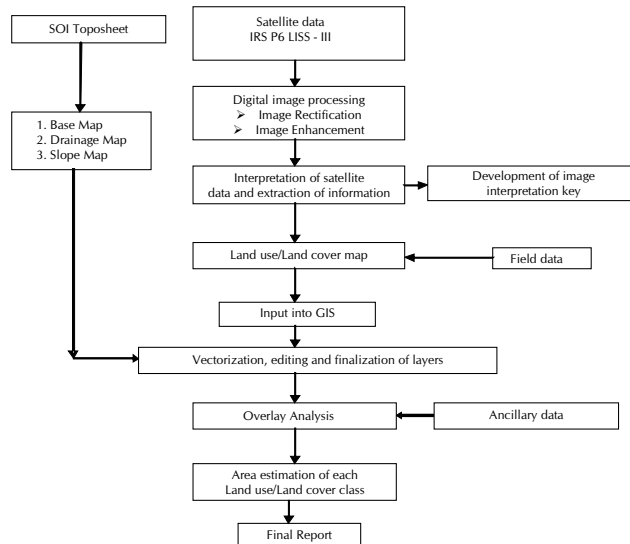


Figure 2: Flow chart showing the broad methodology adopted in the study

Preparation of base map

The base map is the essential requirement for any mapping using remote sensing techniques or by ground methods. It is the map which is used for collection and representing the thematic (resource) information on a common uniform scale. In the study area we used SOI toposheet No. 57 J 03 on 1:50,000 scale for the purpose of base map preparation. The major features like important town or villages, rivers, reserved

Table 1: Image interpretation keys used for land use/ land cover classification

Land use class	Image Characteristics					
	Tone Band 5	Band 7	Band 4, 5 and 7	Texture	Pattern	Location
Built up land	Medium to light	Blue mixed with grey	Fine red and white	Semi circular to circular	Small	In plains Plateaus along the rivers, roads and railways.
Agricultural land	Light to Medium	Red	Fine yellowish	Distinct	Varying size	Plains and valley land
Forest land	Light grey	Light grey	Fine to Medium	Rough	Irregular	High land hills
	Light to Medium	Light red to brown		Irregular with Distinct	Varying irregular size	
Water bodies	Medium Dark grey	Light grey	light brown	rough	irregular	Natural stream
	Dark grey	Medium blue to Dark	Fine	Linear	Long	
Uncultivable land	Very dark grey	Light to dark blue	Fine	Triangular	medium	Surrounded by hills
	Light grey	White to yellowish	Fine	Distinct	Large	
	Medium grey	Brown to grey	Medium to rough	Distinct	Small	

forest area and transport network were traced on tracing sheet on the basis of which the base map has been prepared on 1:50,000 scale.

Preparation of thematic maps

These maps are the true representation of earth's phenomena such as spatial distribution of natural resources existing at the time of survey (Gupta, 2003). In the present study satellite image (IRS-P6 LISS-III) which is a true record of the various environmental resources information on the base map. These maps showing spatial distribution of forest, agriculture, water resources etc., are prepared by visual interpretation of the satellite imagery. Visual interpretation is carried out based on the image characteristics like tone, size, shape, pattern, texture etc. in conjunction with existing map/literature. The image interpretation keys used for land use land cover classification are shown in Table 1. These pre field thematic maps are modified substantiated and confirm after limited field checks.

RESULTS AND DISCUSSION

On the basis of the image interpretation and limited field checks, a total of three thematic maps such as land use/land cover, drainage and slope maps were prepared to study the various land resources in the study area.

Land resources: The land resources are studied by preparing the land use/land cover map. The land use/land cover pattern fall under the broad categories of built up land, agricultural land, forest land, uncultivable land and Waterbodies. The land use and land cover system adopted in mapping of earth surface features is a system derived from the United States Geological Survey (USGS) land use/land cover classification system, showed in Table 2 (Lillisand and Keifer,1979).

Table 2: Showing the LULC classification derived from the USGS classification system

Level 1	Level 2
Urban or built-up land	i.Residential, ii.Industrial, iii.Transportation, iv.Communication, and v.Utilities
Agriculture land	i.Crop land - Wet cultivated land Dry cultivated land ii. Fallow land
Forest land	i.Fairly dense forest land , ii Sparse forest land
Water bodies	i.Streams and canals iiLakes iiTanks or Reservoirs
Uncultivable land	iBarren land ii.Waste land

Builtup land

Residential: Urban or built up land is composed of areas of intensive with much of the land covered by structures. Included in this category are cities, towns, villages, industrial and commercial complexes and institutions. In the study area major towns or villages are Pulivendula, Lingala, Ippatla, Udumukurti, Nallapureddipalli, Dorigallu, Diguvaipalli, Sanivaripalli etc. Classified land use/land cover map of the study area is shown

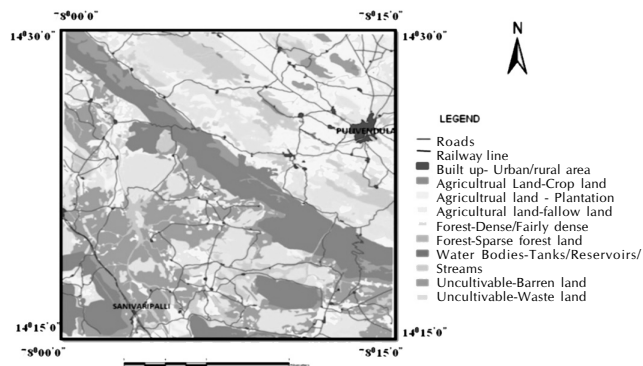


Figure 3: Classified land use/land cover map of the study area

in Fig. 3.

Transportation: The transportation facilities in the study area are roads and railway. The highway roads present in the area are routes between Velidendra-Pulivendula, Jammalamadugu-Pulivendula, Pulivendula-Vempalli, Vemula-Kadiri and Anantapur-Kadiri.

Railway: The unique railway route is South Central railway, Dharmavaram Pakkala branch (Metre gauge) from Dharmavaram to Kalasamudram.

Mining: The industrial mining of asbestos and serpentine minerals is carried out in the villages of Brahmanapalli, Ippatla, Chinna Kuddala, Ramanutlapalli and Lapatnutala which are trending in NW-SE direction.

Agricultural land

All the cultivated land with or without crops orchards and plantations are considered in this class. This land use class is further subdivided into two sub-classes:

Crop land: Crop lands are the agricultural lands under crop. In the study area the crop lands have wet cultivation and dry cultivation. Wet cultivation includes food crops such as paddy, wheat, etc. The areas where the wet cultivation is present are Ippatla, Murarichintala, Kottapalli, Ambakapalli, Udumakuthi, Siddugariipalli etc.

Dry cultivation includes trees orchards, ground nut, etc. and the areas which have this type of cultivation include

Table 3: Bar chart showing land use / land cover statistics of the study area

Description	Area in Ha	Percent to total study area
Built Up area (Urban)	316.13	0.43
Built Up area (Rural)	484.08	0.65
Built Up-Mining / Industrial area	219.02	0.29
Agricultural Land-Crop Land	14820.71	19.43
Agricultural Land-Plantation	2679.10	3.61
Agricultural Land-Fallow-Current Fallow	6089.80	8.85
Forest-(Dry/Moist/Thorn)-Dense/Closed	264.15	0.36
Forest-Scrub Forest	12232.58	16.47
Uncultivable-Barren Rocky/Stony waste land	15708.55	21.16
Uncultivable-Wastelands-Scrub land	18792.10	25.31
Waterbodies-River/Stream/Reservoir/Tanks	2554.3	3.44
Grand Total	74250.46	100.00

Nallapureddipalli, Diguvapalli, Dorigallu and Indukuru.

Fallow land: This refers to all land which was taken up for cultivation but is temporarily out of cultivation for a particular period. The study area have more fallow lands and are surrounded by villages, Krishnamgaripalli, Peddarangapuram-Lingala, Indukuru, 4 km right east of Siddugariipalli and west of Velumvaripalli.

Uncultivable land

Barren land: Barren land covers all lands which are uncultivable like mountains, deserts, bare exposed rock, strip mines, gravel pits and quarries. The barren lands are surrounded by villages Dondi, Diguvapalli and Lapatnutala in North Eastern part and in Southwestern part major land is bare rock exposed land which is not have been used. These lands can be suggested for the industrial and urbanization purposes.

Waste land: Waste lands are non agricultural areas which cannot be brought under cultivation unless at a high cost. In the study area the waste lands occupied nearby villages such as Dondi, Murarichintala, kanampalli and 2 km South of Velumvaripalli.

Forest land

Forest land represents areas that have a tree crown aerial density of 10 percent or more, are stocked with trees capable of producing timber or other wood products and exert an influence on the climate or water regime. In the study area land is classified into two levels.

Fairly dense forest land covered with the hills which are in trending NW-SE adjacent to villages Matnutalpalli, Ambakapalli, Murarichintalapallietc. Sparse forest land located on undulating terrain, up lands, and slopes of the hills. The villages nearer to the sparse forest land are Velumvaripalli, Nakkalapalli, Bandamidapalli etc.

Water resources

The water category includes streams, lakes, canals and reservoirs. To study the water resources of the study area and drainage map is prepared as shown below. The major tanks/reservoirs in the study area present around Lakkasamudram, Reddivaripalli, Sanivaripalli, Gollapalle etc in Southern part of imagery only. Streams, canals are generally located in Northern part of imagery. Maddaleru River is flowing towards NW direction in the Southern Western part of imagery. The villages

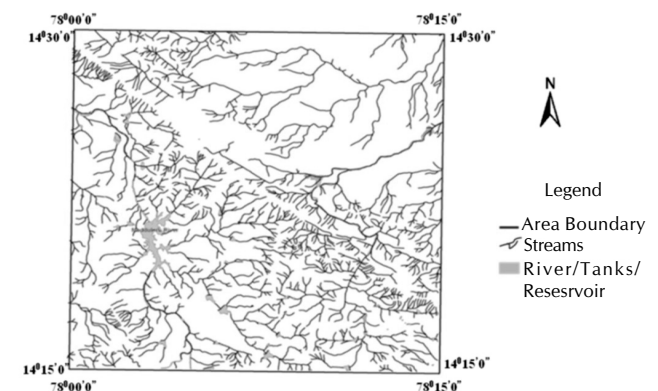


Figure 4: Drainage map of the study area

Table 4: Morphometry of Krishnamgaripalli drainage basin

Morphometric parameters	Stream order (U)			
	I	II	III	IV
Stream number(Nu)	100	20	6	1
Stream length(L_u) km	98.75	33.25	29.75	10.25
Bifurcation ratio (R_b) = $Nu/Nu+1$	4.76	2.85	3	-
Stream length ratio $R_l = L_u/Lu-1$	-	1.85	1.66	2.06
Total drainage area (Au)	332.5 km ²			
Average Bifurcation ratio (R_b)	3.6695			
Drainage density, $D_d = \frac{1}{A} L_u/Au$	162.95/332.5 = 0.490 km/km ²			
Drainage Texture	0.49 km/ km ² . (<2), Very coarse drainage texture			

adjacent to the river are Sanivaripalli, Adavibrahmanpalli, Nakkalapalli, Dorigallu, Indukuru and Maartadu.

Watershed modeling: After studying the total conditions of the area it is concluded that, the region is very poor in water resources, top priority should be given for its development. The ground water distribution in the study area is not uniform and it depends on the drainage pattern. Drainage pattern of any terrain reflects the characteristic of surface as well as subsurface formations in a watershed area (Horton, 1932). For this watershed, near the village Krishnamgaripalli is demarcated so as to fulfill the needs. The watershed area is represented between Eguvapalli in the west and Krishnamgaripalli in the east. The demarcated watershed is named after Krishnamgaripalli village.

The watershed covers an area of 332.5 km² (Fig. 4). It enjoys a tropical climate with rainfall between 100-150cm annually. A watershed represents all of the stream tributaries that flow to some location along the stream channel. Watershed is a basic unit for water resource survey. Land cover parameters are valuable in these studies as direct inputs and also as indirect inputs by inferring basin characteristics such as drainage area, drainage density, drainage texture, bifurcation ratio etc., (Montgomery, 2006). By the morphometric analysis all the linear aspects and drainage aspects of the watershed is calculated.

CONCLUSIONS

From the above discussions it is concluded that Remote sensing data provide reliable, accurate, baseline information for land use / land cover mapping. Vectorisation, editing, overlay analysis and area estimation of land use classes is easier in Geographic Information System. The present study reveals that in the study area the land use-land cover pattern fall under the broad categories of built-up land, agricultural land, forest land and waste land. In the study area the uncultivable barren and waste land covers an area of 48 per cent of land, which is unused for neither cultivation nor settlements. The important point to be considered in this study is, there is 0.36 per cent of dense forest and 16.47 per cent of scrub forest. i.e., less than 17 per cent. The area covered by forest is drastically decreasing in the last few years because people use this forest for fuel, fodder purposes and over grazing of the animals. Therefore fencing of the forest boundary is much needed. Agro forestry system has been suggested on hilly terraces, and some waste land areas to meet the needs of local people for fuel and fodder. In the study area nearly 50 per cent of the villages are without pucca roads and the situation is worst in

the case of communication facilities like post office, telegraph office etc. So provision of basic infrastructure for socio economic development is much needed. The ground water distribution in the study area is not uniform and it depends on the drainage pattern. Since the region is very poor in water resources, top priority should be given for its development. For this watershed, near the village Krishnamgaripalli is demarcated so as to fulfill the needs. The barren land in the study area can be used for industrialisation and urban planning.

ACKNOWLEDGEMENT

The authors are grateful to APCOST, Hyderabad for financial support in the form of student project fellowship. The authors are thankful to Dr. K. Mruthyunjaya Reddy, Director General, APSRAC, Hyderabad for his kind permission to undertake this study in lab. The authors are thankful to Sri I.V Reddy, Professor, Department of Geology and Geoinformatics, Yogi Vemana University, Kadapa, A.P for his guidance, critical discussions and constructive suggestions during the interpretation of the satellite data and also in the field.

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